PROJECT DESIGN PHASE

# Problem–Solution Fit

Script-Controlled ACL – Restrict Record Access Based on Field Value

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| Problem statement | Script-Controlled ACL – Restrict Record Access Based on Field Value |
| Team ID | SWTID-2026-8933 |
| Date | 1 October 2026 |
| Maximum Marks | 2 marks |

## 1. Problem–Solution Fit

Problem–Solution Fit means identifying a real problem experienced by a target user and checking whether the proposed solution addresses that problem. For this project, the fit is evaluated around controlled access to ServiceNow records when access depends on a field value.

## 2. Problem Statement

In a ServiceNow application, users may need access to records based on the value of a particular field (for example, a department, assignment group, ownership flag, or status). A broad role-based or table-level permission may not express that record-by-record rule. Without a field-sensitive access check, users could see records outside the intended scope, while legitimate users may also be blocked if access rules are too restrictive or unclear.

## 3. Target Users and Stakeholders

| Stakeholder | Need / concern |
|---|---|
| ServiceNow administrators / developers | Configure and maintain access rules that match business requirements and are testable. |
| Employees / fulfiller users | See and work with records they are authorized to access, without unnecessary access to other records. |
| Application or data owners | Protect sensitive or operational records and apply consistent access boundaries. |
| Auditors / security reviewers | Understand the access rule and verify that both allowed and denied cases behave as intended. |

## 4. Proposed Solution

Create a script-controlled Access Control List (ACL) rule that evaluates the relevant field value when a user attempts to access a record. The rule should grant access only when the configured condition is satisfied and deny access when it is not. The exact table, field, roles, and condition must be finalized from the project's business requirements.

### Expected behavior

- Authorized user + matching field value: access is granted, subject to any other applicable ACL checks.
- User does not meet the condition: access is denied.
- Missing, unexpected, or invalid field value: access follows a defined deny-by-default rule unless the requirements explicitly specify otherwise.

## 5. Problem–Solution Fit Canvas

| Canvas element | Project-specific entry |
|---|---|
| Customer / user | ServiceNow users who need record access, plus administrators responsible for enforcing access policy. |
| User problem | Users need access limited by a record's field value; general access settings may not represent this requirement precisely. |
| Current workaround / alternative | Possible alternatives include role-based ACLs, conditions, or manual process controls. The actual current method should be confirmed with the project owner. |
| Proposed solution | A scripted ACL that checks the target field and permits access only when the business condition is met. |
| Value / benefit | More precise record-level access, consistent enforcement, reduced unintended visibility, and a rule that can be tested and maintained. |
| Key assumptions | The relevant field is populated reliably; the business rule is unambiguous; the ACL is applied to the correct table and operation; required roles and exceptions are known. |
| Risks / constraints | Incorrect field or condition, incomplete ACL coverage, conflicts with other ACLs, performance impact from complex scripts, and insufficient test coverage. |
| Success measures | Positive tests allow intended users; negative tests deny unintended users; read/write/create/delete behavior matches requirements; no unrelated user workflow is broken. |

## 6. Validation Plan

The solution is considered to fit the problem when tests demonstrate that access follows the approved field-based rule for both permitted and prohibited cases. Complete the following checks in a non-production instance before deployment:

| Test case | Expected result | Status |
|---|---|---|
| User meets rule and field value matches | Access granted, if no other ACL blocks it | Pending |
| User does not meet rule | Access denied | Pending |
| Field value differs from allowed value | Access denied | Pending |
| Field is empty or unexpected | Result follows documented fallback; preferably deny | Pending |
| Check each required operation (read/write/create/delete) | Only explicitly permitted operations succeed | Pending |
| Regression test for unaffected users and records | Existing authorized workflows continue to work | Pending |

## 7. Conclusion

The proposed scripted ACL is intended to address the need for field-value-based record access in ServiceNow. Problem–solution fit should be confirmed by reviewing the actual business rule with stakeholders and completing the validation cases above. This document describes the proposed fit; it does not claim that the ACL has already been implemented or tested.

## 8. References

1. Supplied sample: Project Design Phase – Problem–Solution Fit Template (1 page; includes the purpose of problem–solution fit and a template heading).

2. ServiceNow table, field, role, and ACL details: to be added from the project implementation and approved requirements.

*Note: Team ID and project-specific table/field/business-condition details are intentionally left for the project team to confirm.*